# Karan M

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## Summary

B.Tech graduate in Artificial Intelligence and Data Science with a strong foundation in Python, machine learning, OOP, and DBMS. Built production-ready ML projects incorporating explainable AI techniques (SHAP, LIME) and React-based web applications. Eager to contribute to data-driven teams and grow in a structured engineering environment.

## Experience

Online Intern - Data Science and Data Analytics, Coimbatore Institute of Technology, Jun-Jul 2024
Completed an intensive data science internship focused on data analysis workflows and ML fundamentals. Worked on structured datasets applying preprocessing, exploratory data analysis, and predictive modelling techniques using Python and pandas.

## Projects

**Ensemble AI-based crop recommendation system**

- Implemented the Ensemble AI-based Sustainable Crop Planning Framework that recommends suitable crops based on soil and environmental conditions using **6 ML models, Explainable AI(XAI)**. This project gave me hands-on experience in building an end-to-end machine learning pipeline — from raw data to explainable predictions.

**AI Resume Screener**

- Built an AI-powered resume screening tool using **Python, NLP techniques, Streamlit** and Sentence-BERT for semantic similarity matching. Developed weighted scoring engine **semantic, skills, experience, education** with interactive **Streamlit** dashboard featuring radar charts and skill gap heatmap.

**Fleet Service Management System**

- Designed an **OOP**-based fleet management system modeling vehicle service lifecycles, using abstract base classes, inheritance (type-specific service logic), and the Strategy pattern for pluggable service-tier pricing using **Python**. Built a normalized **SQLite** schema with foreign-key constraints and CTE-based queries to identify service-overdue vehicles and technician workload distribution.

## Skills

**Dev:** Python, Java, SQL

**ML/AI:** scikit-learn, pandas, numpy, explainability (SHAP/LIME), NLP

**CS Fundamentals:** Data Structures, AI Algorithms, Database Management Systems (DBMS), Object-Oriented Programming (OOP)

**Soft skills:** Problem-Solving, Team Collaboration, Quick Learning, Time Management

## Certifications

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|---|---|
| *Google AI Essentials* | *Coursera* |
| *Introduction to Model Context Protocol* | *Anthropic* |
| *Project Management Job Simulation* | *Accenture* |
| *Google Prompting Essentials* | *Coursera* |

## Education

**Sri Krishna College of Engineering and Technology** *Sept 2022 - March 2026*
*B.Tech. - Artificial Intelligence and Data Science*

- CGPA: 7.67 / 10

**CIT Sandwich Polytechnic College, Coimbatore** *Jan 2023*
*Diploma - Mechanical Engineering (Sandwich)*

- Percentage: 92 / 100

**Bharathi.Matriculation. Higher Secondary.School, Coimbatore** *Mar 2019*
*10th — State Board of School Examination*

- Percentage: 83.20 / 100